

Cambridge (CIE) IGCSE Chemistry
Paper 6: Alternative to Practical (Set D)

Saturday 26 April 2025

Morning (Time: 1 hour 0 minutes)

Total marks

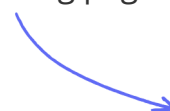
/ 40

Instructions

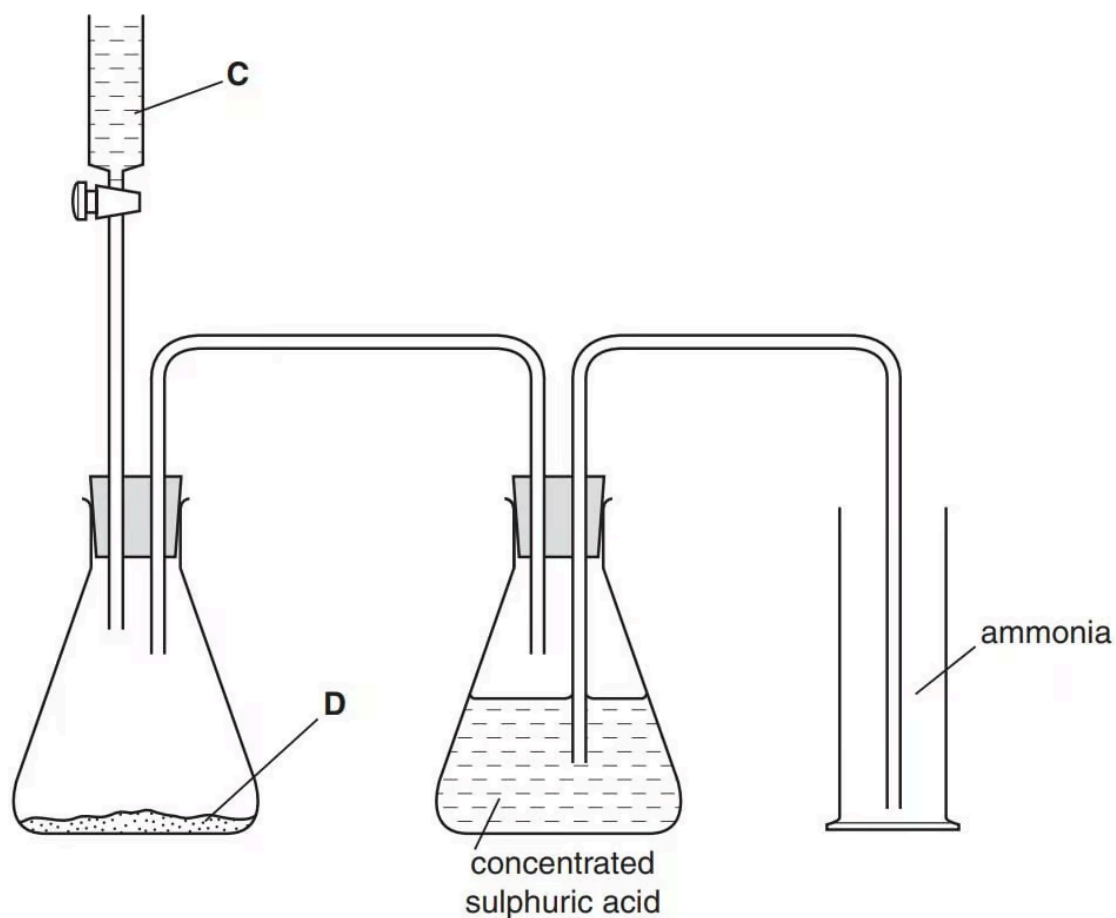
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- You should show all your working and use appropriate units.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 (a) Ammonia is produced when aqueous sodium hydroxide is warmed with ammonium sulfate. Ammonia is less dense than air and very soluble in water. The apparatus below was used to prepare a sample of dry ammonia gas.



Name substance **C**.

(1 mark)

(b) Name substance **D**.

(1 mark)

(c) What necessary piece of equipment is missing in the diagram?

(1 mark)

(d) There are two other mistakes in the apparatus shown in the diagram. Identify and explain these mistakes.

mistake 1

explanation

mistake 2

explanation

.....

.....

.....

.....

(4 marks)

- 2 (a)** A solid **U** was analysed. **U** was a soluble metal sulfate. The tests on **U**, and some of the observations are in the following table.

Complete the observations.

Tests	Observations
Test 1: Appearance of solid U .	Pink crystals
Test 2: Solid U was heated gently and then strongly in a test-tube.	condensation droplets formed on the sides of the test-tube
Solid U was added to distilled water in a test-tube and shaken until dissolved.	
Test 3 part (i): The solution was divided into three equal portions in separate test-tubes and the following tests carried out. Several drops of aqueous sodium hydroxide were added to the first portion of the solution and the test-tube shaken.	pale brown precipitate
Test 3 part(ii): Then hydrogen peroxide solution was added to the mixture and the gas given off was tested.	effervescence glowing splint re-lit
Test 4: Dilute nitric acid was added to the second portion of the solution followed by barium nitrate solution.	
Test 5: Dilute nitric acid was added to the third portion of the solution followed by silver nitrate solution.	

- i) State the expected observations for **Test 4**.

[2]

ii) State the expected observations for **Test 5**.

[1]

.....

.....

.....

(3 marks)

(b) What does **Test 5** tell you about solid **U**?

.....

(1 mark)

(c) Name the gas given off in **Test 3**.

.....

(1 mark)

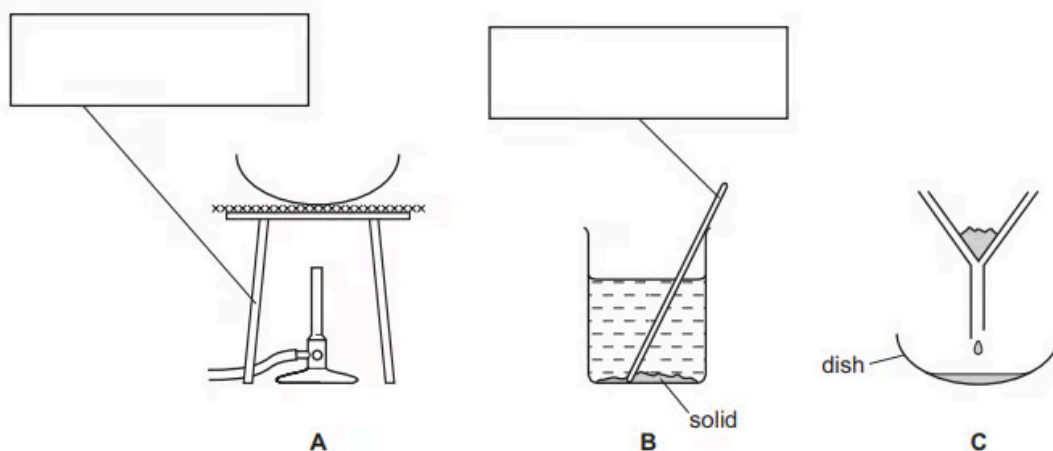
(d) What conclusions can you draw about solid **U**?

.....

.....

(2 marks)

- 3 (a)** The diagrams show the apparatus used to obtain crystals of calcium chloride from a mixture of solid calcium chloride and solid calcium carbonate. Calcium chloride is soluble in water and calcium carbonate is insoluble in water.



Complete the boxes to name the apparatus.

(2 marks)

- (b)** i) Write down the order in which the apparatus should be used in this experiment.

[1]

- ii) Name the separation process in **C**.

[1]

(2 marks)

(c) i) What has been added to the mixture in **B**?

[1]

ii) What is the general name given to the liquid in the dish in **C**?

[1]

.....

.....

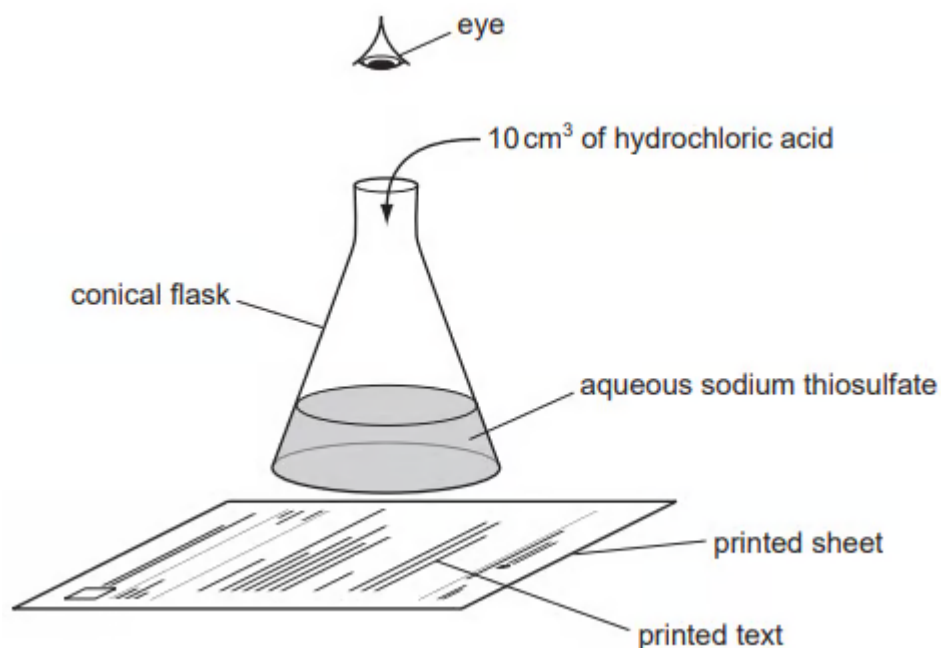
(2 marks)

(d) How would you know when to stop heating the dish in **A**?

.....

(1 mark)

- 4 (a) Hydrochloric acid reacts with aqueous sodium thiosulfate to form a precipitate, which makes the solution turn cloudy. The formation of the precipitate can be used to show how fast the reaction proceeds, using the apparatus shown below.



A student used this method to investigate the effect of changing the concentration of the sodium thiosulfate solution on the speed of the reaction. The student used different concentrations of sodium thiosulfate solution. All other variables were kept the same.

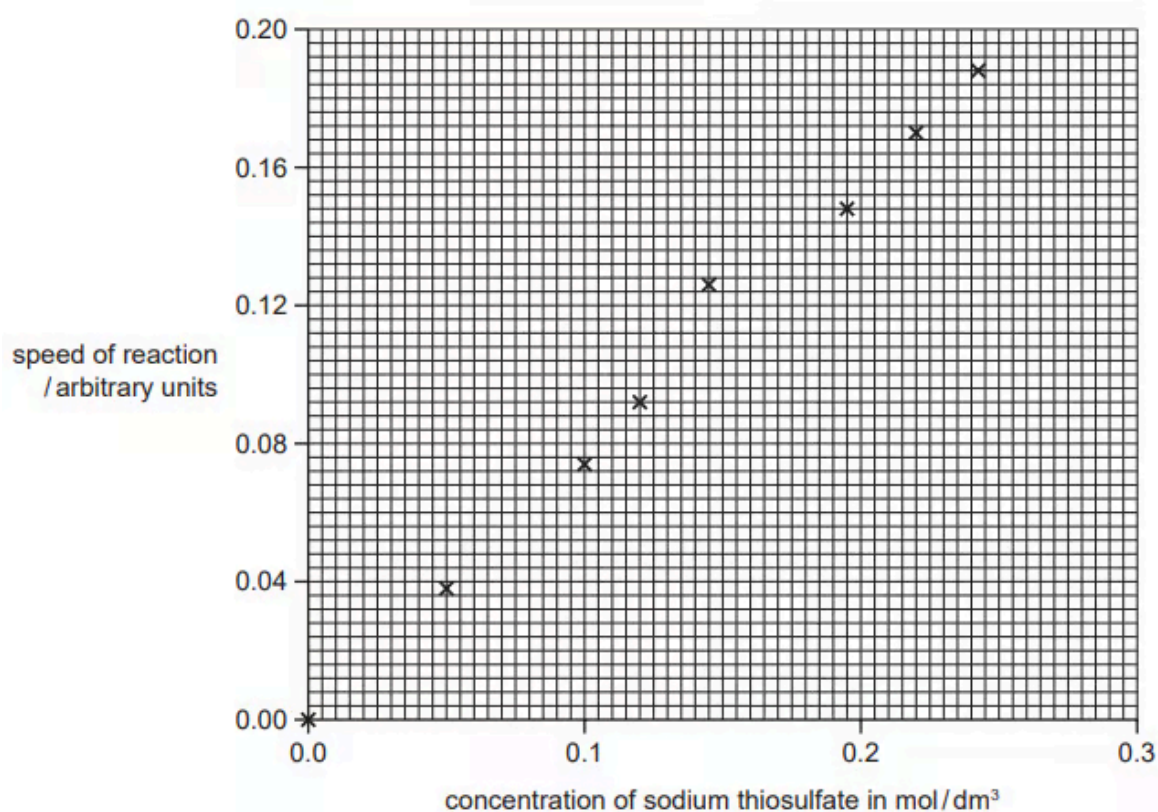
Give **two** variables which were kept the same in the investigation.

.....

.....

(2 marks)

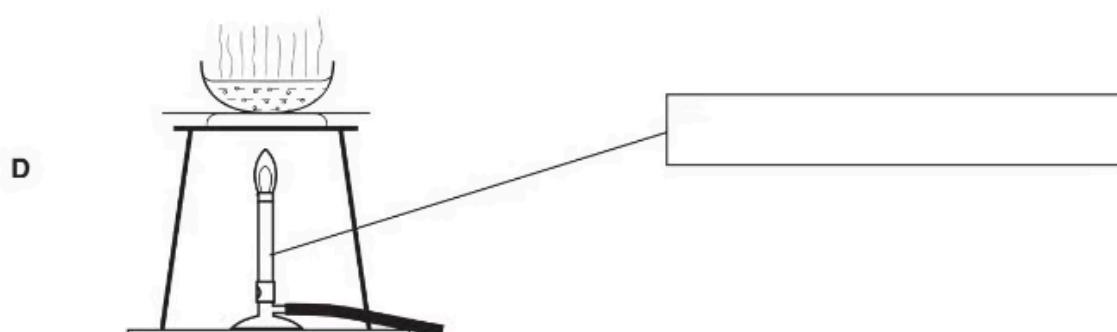
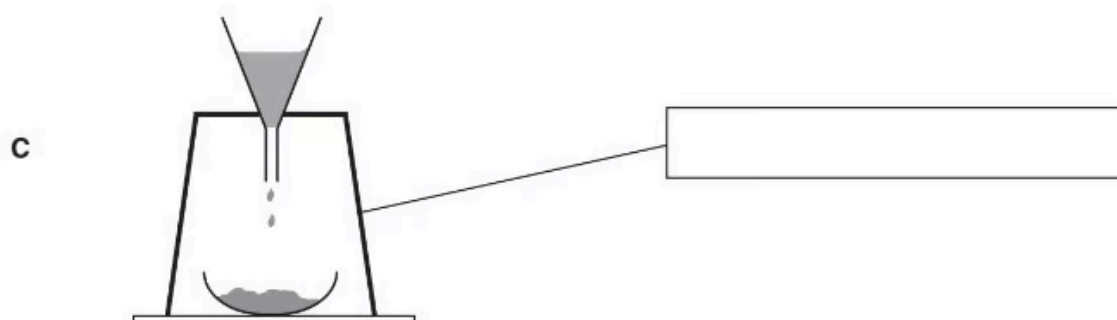
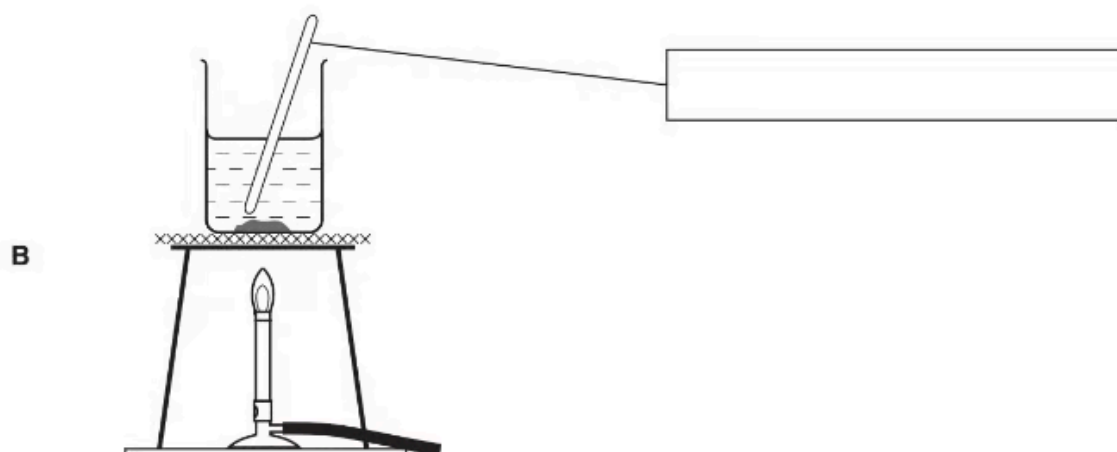
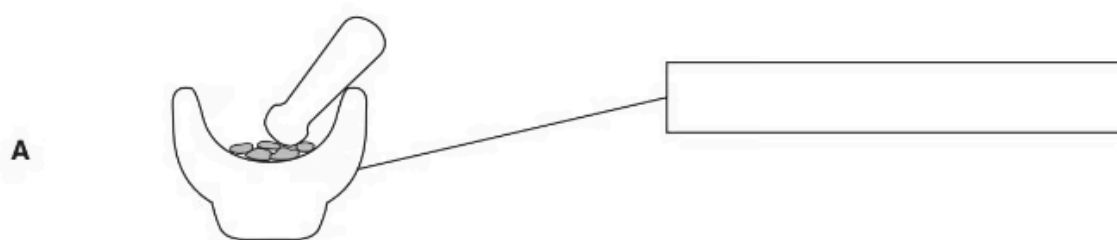
(b) The results of the experiments are shown plotted on the grid below



Draw a line of best fit on the grid.

(1 mark)

5 (a) Look at the diagrams of common laboratory apparatus.



Complete the empty boxes to identify the pieces of apparatus labelled.

(4 marks)

(b) What name is given to the separation method in **C**?

(1 mark)

(c) Which apparatus would be most suitable to obtain crystals from an aqueous solution of copper(II) sulphate?

(1 mark)

- 6 (a)** A mixture of two solids, **E** and **F**, was analysed. Solid **E** was the water-soluble salt aluminium chloride, AlCl_3 , and solid **F** was an insoluble salt. The tests on the mixture and some of the observations are in the following tables. Complete the observations in the tables.

tests	observations
Distilled water was added to the mixture in a boiling tube. The contents of the boiling tube were shaken and filtered, keeping the filtrate and residue for the following tests.	
<u>tests on the filtrate</u> The filtrate was divided into five portions in five test-tubes. The first portion was used to describe the appearance of the filtrate	appearance

(1 mark)

(b)

tests	observations
Several drops of aqueous sodium hydroxide were added to the second portion of the solution. Excess aqueous sodium hydroxide was then added to the test-tube.	

(3 marks)

(c)	tests	observations
	Aqueous ammonia was added to the third portion, dropwise and then in excess.	

(2 marks)

(d)	tests	observations
	To the fourth portion of the solution, dilute nitric acid and aqueous silver nitrate were added.	

(2 marks)

(e)	tests	observations
	<u>tests on the residue</u> (i) To a little of the residue, dilute hydrochloric acid was added. The gas given off was tested. (ii) The residue was heated, gently then strongly.	rapid effervescence gas turned limewater milky solid changed colour from green to black

What conclusions can you draw about solid **F**?

(2 marks)